

ABHINAV KAUSHIK

Software Engineer - Internal AI

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SUMMARY

Senior AI engineer with 8+ years shipping production software and applied AI across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI. Builds agentic workflows and RAG pipelines with LLM APIs and LangGraph in Python, owns them end to end with FastAPI, Docker and CI/CD, and keeps them reliable with automated evaluation, tracing and monitoring. Outside work, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | United Health Group Bengaluru, India | Nov 2025 - Present

- Shipped a production lab-extraction agent end to end with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents and automating the work behind 800 Jira tickets.
- Engineered a production multi-agent chatbot in LangGraph that answers internal business questions with PySpark analysis, multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Built an internal mock-generator tool producing 36+ discipline-specific UI mockups from Jira stories and requirements with HTML, Next.js and Claude 4.6, speeding up prototyping and stakeholder review.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents with LLM evaluators, tracing and failure detection, catching quality drops and reducing manual review by 43%.
- Fine-tuned a cross-encoder on positive/negative pairs to re-rank AARP survey retrieval, using Databricks notebooks and Vector Search for embeddings and retrieval, improving downstream question-answering accuracy.

Senior Machine Learning Engineer | Viavi Solutions Chennai, India | Sep 2024 - Nov 2025

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built and deployed an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients across NAM, EMEA and LATAM from client requirements, with 7 moving into production or paid client work.
- Built CI/CD pipelines for service and model deployment with Jenkins, Bitbucket and Docker, adding unit and integration tests and automated checks so releases stayed repeatable and reviewed.

Data Scientist | Viavi Solutions Bengaluru, India | Nov 2021 - Sep 2024

- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Contributed to network-device inventory demand forecasting using predictive models to improve planning and reduce revenue-at-risk; targeted a conservative 10% reduction in excess/at-risk inventory and higher ARPU.
- Trained and deployed a production time-series forecasting framework combining LSTM and XGBoost to model call quality, SMS delivery rates and network efficiency metrics for the Delta cellular network.
- Engineered a unified time-series data pipeline in SQL and Python to ingest call logs, SMS delivery statistics and signal metrics across multiple time intervals, joined by user ID and network location for downstream forecasting.

Data Analyst | Optum Gurugram, India | Jun 2021 - Nov 2021

- Optimized a scikit-learn Random Forest classifier for member risk stratification with 12+ engineered features, GridSearchCV and 5-fold CV, lifting AUC from 0.74 to 0.81 and recall from 0.62 to 0.73.
- Optimized hospital staff scheduling in Python as a mixed-integer linear program with shift-coverage constraints, and modelled patient wait times with queueing statistics to balance staffing levels against demand.
- Analyzed member plan utilization and renewal behavior across individual and child plan segments using BigQuery, building segmentation queries and reporting pipelines that fed downstream analytics and member risk modelling.

Data Engineer | Nippon Data Systems Gurugram, India | Nov 2019 - May 2021

- Engineered SQL, PySpark and Airflow pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Built customer dashboards for financial and product insights using QlikSense, SQL and SAP HANA, increasing user engagement by 40%.
- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach.

- Wrote complex SAP HANA SQL with CTEs, joins, subqueries, window functions and aggregations to extract and validate financial modelling data.
- Built reusable SQL data-preparation workflows across financial and product datasets, refining queries and filters to surface relevant signals for financial teams.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Internal Operations Agent Platform

- Built an internal agent platform that automates cross-functional back-office workflows, combining LangGraph orchestration, retrieval over internal documents and tool connectors exposed through MCP.
- Defined done with secure auth, error handling and rollback, evaluated each agent with an automated eval harness, and ran the services in containers with monitoring.
- **Technologies:** Python, LangGraph, MCP, FastAPI, Pydantic, Docker, CI/CD, pgvector

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection | Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 - Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning, A/B Testing

Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, CI/CD, GCP, Azure, Vercel

Data Platforms: Databricks, Databricks SQL, Databricks Vector Search, BigQuery, Apache Spark, Supabase

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

Search: Apache Solr, Lucene Indexing, Information Retrieval

BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native
German: A1, Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies